

Reproducibility Supplement

Innovation-Normalized Conformal Prediction for Autoregressive Episodes

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1 Software and random-number configuration

The comparison experiments used NumPy 2.2.6, SciPy 1.15.3, scikit-learn 1.6.1 and sklearn-quantile 0.1.1. Separate spawned random streams generated training, calibration, evaluation and model randomness. The master seeds were 2026092712 for the common-dynamics confirmation experiments and 2026092714 for the misspecification experiments. The fitted-mean validation used seed 2026092917. These integer identifiers are provided to reproduce the random streams; they have no mathematical interpretation.

The reusable IN-ARCP package implements the coefficient fit, history scale, rank calibration and interval inversion. It also provides the known-coefficient expected length and leading asymptotic efficiency terms. Its source license is MIT. The package example and tests are distinct from the full comparative simulation runs.

2 Ad-EffOrrt implementation

The linear optimization uses 1000 iterations, steps proportional to $t^{-0.6}$ and smoothing parameter 0.1. Its residual quantile forest has 100 trees, maximum depth five and quantile level 0.9. The sequential accumulation order was compared with the authors' implementation. The relevant source revision and file hash are recorded in `reproducibility/comparator_provenance_check.json`. The scale-invariant adaptation normalizes each complete episode by the RMS of its history before fitting and calibration, and rescales predicted endpoints afterwards. It is distinct from the authors' original procedure.

3 Numerical integration and Monte Carlo validation

The complete mean-length calculation uses $m = 2$, $\rho = 0.4$, $N = 2$, training scales $(0.7, 1.3)$, $n = 19$, $\alpha = 0.1$ and unit test scale. The clipping threshold is 0.98. Circular integration uses 2048 midpoint angles; the beta expectation uses 64-point Gauss–Jacobi quadrature. Stieltjes integration retains both endpoint masses. Refinement uses 32, 64 and 128 bins. The full program and machine-readable output are included as `reproducibility/recheck_mathematics.py` and `reproducibility/mathematics_recheck.json`.

The 128-bin calculation gives mean length 6.9372263. Monte Carlo simulation over 30000 fitted repetitions, each evaluated on 32 test episodes, gives 6.9425058 with standard error 0.0257901. The discrepancy is 0.205 standard errors. The corresponding estimated coverage is 0.9001042 with standard error 0.0004800. Increasing from 64 to 128 bins changes the mean by 0.0006748. These results evaluate one fixed-training-scale configuration; they do not assess integration over a random training-scale distribution.

Additional component checks evaluate 36 score-CDF configurations, six conditional mean configurations, 32 training-CDF points and four clipping probabilities. Matrix identities are

evaluated at five history lengths and five coefficients. These computations assess implementation accuracy at the specified inputs. The mathematical justification remains the derivations in the article.

4 Result files and reproduction scope

The `supplementary_results` directory contains all saved method-level and paired results from the main scenarios, invariant adaptations, dynamics challenges and Student-reference comparisons. The overview figure selects the common Gaussian AR(1) scenarios from these records; the CSV files retain the other evaluated scenarios and all recorded gains and losses. Comparator configurations include both published constructions and explicitly identified adaptations or model-specific implementations.

The manuscript's figure builder reads preserved simulation summaries; it does not refit comparators. The accompanying source bundle includes these summaries, the figure-generation program and a hash manifest. The Overleaf archive contains the files needed to compile the article and this supplement. The open-source predictor release does not contain the complete comparator simulator collection. Consequently the predictor release, figure-reproduction bundle and manuscript archive serve different reproducibility purposes.